

Introduction to Linear Algebra
Lecture Notes

Introduction - What is Linear Algebra?

It is well-known that the total cost of a purchase of amounts g_1, g_2, g_3 of some goods at prices p_1, p_2, p_3 , respectively, is an expression

$$p_1g_1 + p_2g_2 + p_3g_3 = \sum_{i=1}^3 p_i g_i.$$

Expressions of this kind,

$$a_1x_1 + \cdots + a_nx_n$$

are called **linear forms in variables** $x_1, \dots, x_n$ **with coefficients** $a_1, \dots, a_n$.

- Linear Algebra studies the mathematics of linear forms.
- Over the course, we shall develop increasingly compact notation for operations of Linear Algebra. In particular, we shall discover that

$$p_1g_1 + p_2g_2 + p_3g_3$$

can be very conveniently written as

$$\begin{bmatrix} p_1 & p_2 & p_3 \end{bmatrix} \begin{bmatrix} g_1 \\ g_2 \\ g_3 \end{bmatrix}$$

and then abbreviated

$$\begin{bmatrix} p_1 & p_2 & p_3 \end{bmatrix} \begin{bmatrix} g_1 \\ g_2 \\ g_3 \end{bmatrix} = P^T G,$$

where

$$P = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} \quad \text{and} \quad G = \begin{bmatrix} g_1 \\ g_2 \\ g_3 \end{bmatrix}$$

are vectors and T (transposition) turns row vectors into column vectors, and vice

versa:

$$P^T = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix}^T = \begin{bmatrix} p_1 & p_2 & p_3 \end{bmatrix}, \quad \text{and} \quad \begin{bmatrix} p_1 & p_2 & p_3 \end{bmatrix}^T = \begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix}.$$

- Unlike, say, Calculus, Linear Algebra focuses more on the development of a special mathematics language rather than on procedures.

1 Linear Equations in Linear Algebra

1.1 Systems of linear equations

A **linear equation** in the variables $x_1, \dots, x_n$ is an equation that can be written in the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

where b and the **coefficients** $a_1, \dots, a_n$ are real numbers. The subscript n can be any natural number.

A **system of simultaneous linear equations** is a collection of one or more linear equations involving the same variables, say $x_1, \dots, x_n$. For example,

$$\begin{aligned} x_1 + x_2 &= 3 \\ x_1 - x_2 &= 1 \end{aligned}$$

We shall abbreviate the words “**a system of simultaneous linear equations**” just to “**a linear system**”.

A **solution** of the system is a list $(s_1, \dots, s_n)$ of numbers that makes each equation a true identity when the values $s_1, \dots, s_n$ are substituted for $x_1, \dots, x_n$, respectively. For example, in the system above $(2, 1)$ is a solution.

The set of all possible solutions is called the **solution set** of the linear system.

Two linear systems are **equivalent** if they have the same solution set.

Elementary Operations: We shall use the following elementary operations on systems of simultaneous linear equations:

- **Replacement:** Replace one equation by the sum of itself and a multiple of another equation.

- **Interchange:** Interchange two equations.
- **Scaling:** Multiply all terms in a equation by a nonzero constant.

Note: The elementary operations are reversible.

Theorem 1.1 (Elementary operations preserve solutions.). If a system of simultaneous linear equations is obtained from another system by elementary operations, then the two systems have the same solution set.

We shall prove later that a system of linear equations has either

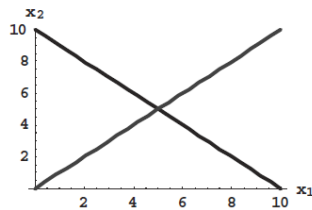
- no solution, or
- exactly one solution, or
- infinitely many solutions,

under the assumption that the coefficients and solutions of the systems are *real or complex numbers*.

A system of linear equations is said to be **consistent** if it has solutions (either one or infinitely many), and a system is **inconsistent** if it has no solution.

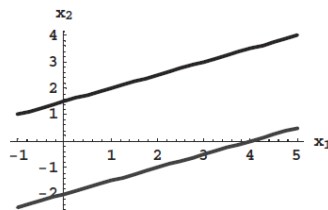
Example 1.2 (Two equations in two variables).

$$\begin{aligned} x_1 + x_2 &= 10 \\ -x_1 + x_2 &= 0 \end{aligned}$$



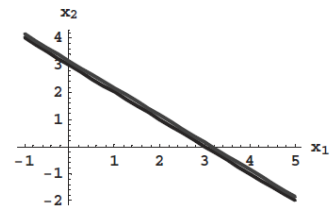
one unique solution

$$\begin{aligned} x_1 - 2x_2 &= -3 \\ 2x_1 - 4x_2 &= 8 \end{aligned}$$



no solution

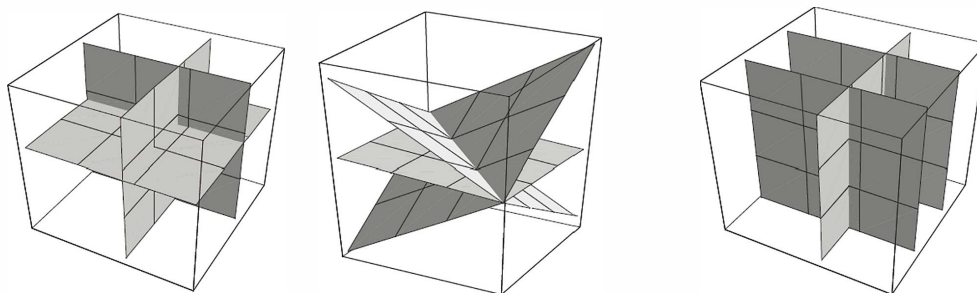
$$\begin{aligned} x_1 + x_2 &= 3 \\ -2x_1 - 2x_2 &= -6 \end{aligned}$$



infinitely many solutions

Example 1.3. Three equations in three variables Each equation determines a plane in 3-space.

- i) The planes intersect in one point. (*one solution*) ii) The planes intersect in one line. (*infinitely many solutions*) iii) There is not point in common to all three planes. (*no solution*)



1.1.1 Solving a linear system

The basic strategy is

to replace one system with an equivalent system (that is, with the same solution set) which is easier to solve.

Existence and uniqueness questions

- Is the system consistent?
- If a solution exist, is it *unique*?

Equivalence of linear systems

- When are two linear systems equivalent?

Checking solutions Given a solution of the system of linear equations, how to check that it is correct?

1.2 Row reduction and echelon forms

Matrix notation

It is convenient to write coefficients of a linear system in the form of a matrix, a rectangular table. For example, the system

$$\begin{aligned}x_1 - 2x_2 + 3x_3 &= 1 \\x_1 + x_2 &= 2 \\x_2 + x_3 &= 3\end{aligned}$$

has the **matrix of coefficients**

$$\begin{bmatrix} 1 & -2 & 3 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

and the **augmented matrix**

$$\begin{bmatrix} 1 & -2 & 3 & 1 \\ 1 & 1 & 0 & 2 \\ 0 & 1 & 1 & 3 \end{bmatrix};$$

notice how the coefficients are aligned in columns, and how missing coefficients are replaced by 0.

The augmented matrix in the example above has 3 rows and 4 columns; we say that it is a 3×4 matrix.

Definition 1.4. Consider linear system of m equations with n variables

$$\begin{aligned}a_{11}x_1 + \cdots + a_{1n}x_n &= b_1 \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n &= b_m\end{aligned}$$

A matrix of the coefficients of the system above is an $\mathbf{m} \times \mathbf{n}$ **matrix**

$$M_{m,n} \equiv \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}$$

Similarly, for the augmented matrix we add a right hand side vector $b = (b_1, \dots, b_m)^T$

$$\widetilde{M}_{m,n+1} \equiv [M_{m,n} | b] = \left[\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ \vdots & & \vdots & \vdots \\ a_{m1} & \cdots & a_{mn} & b_m \end{array} \right]$$

Elementary row operations:

- **Replacement:** Replace one row by the sum of itself and a multiple of another row.
- **Interchange:** Interchange two rows.
- **Scaling** Multiply all entries in a row by a nonzero constant.

Theorem 1.5. The two matrices are **row equivalent** if there is a sequence of elementary row operations that transforms one matrix into the other.

Note:

- The row operations are reversible.
- Row equivalence of matrices is an equivalence relation on the set of matrices.

Theorem 1.6 (Row Equivalence). If the augmented matrices of two linear systems are row equivalent, then the two systems have the same solution set.

A **nonzero** row or column of a matrix is a row or column which contains at least one nonzero entry.

We can now formulate a theorem (to be proven later).

Theorem 1.7 (Equivalence of linear systems). Two linear systems are equivalent if and only if the augmented matrix of one of them can be obtained from the augmented matrix of another system by row operations and insertion / deletion of zero rows.

A **nonzero** row or column of a matrix is a row or column which contains at least one nonzero entry. A **leading entry** of a row is the leftmost nonzero entry (in a non-zero row).

Definition 1.8 (Row Echelon Form). A matrix is in **(row) echelon form** if it has the following three properties:

1. All nonzero rows are above any row of zeroes.
2. Each leading entry of a row is in column to the right of the leading entry of the row above it.

3. All entries in a column below a leading entry are zeroes.

If, in addition, the following two conditions are satisfied,

4. All leading entries are equal 1.
5. Each leading 1 is the only non-zero entry in its column

then the matrix is in **reduced echelon form**.

An **echelon matrix** is a matrix in echelon form.

Any non-zero matrix can be **row reduced** (that, transformed by elementary row operations) into a matrix in echelon form (but the same matrix can give rise to different echelon forms).

Definition 1.9. The following is a schematic presentation of an echelon matrix:

$$\begin{bmatrix} \blacksquare & * & * & * & * \\ 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & \blacksquare & * \end{bmatrix}$$

and this is a reduced echelon matrix:

$$\begin{bmatrix} 1 & 0 & * & 0 & * \\ 0 & 1 & * & 0 & * \\ 0 & 0 & 0 & 1 & * \end{bmatrix}$$

Theorem 1.10 (Uniqueness of the reduced echelon form). Each matrix is row equivalent to one and only one reduced echelon form.

Definition 1.11. A **pivot position** in a matrix A is a location in A that corresponds to a leading 1 in the reduced echelon form of A . A **pivot column** is a column of A that contains a pivot position.

Example 1.12 (The Row Reduction Algorithm).

$$\begin{bmatrix} 0 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 1 & 2 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Definition 1.13. A **pivot** is a nonzero number in a pivot position which is used to create zeroes in the column below it.

Definition 1.14 (A rule for row reduction). $\square$

1. Pick the leftmost non-zero column and in it the topmost nonzero entry; it is a **pivot**.
2. Using **scaling**, make the pivot equal 1.
3. Using **replacement** row operations, kill all non-zero entries in the column below the pivot.
4. Mark the row and column containing the pivot as **pivoted**.
5. Repeat the same with the matrix made of not pivoted yet rows and columns.
6. When this is over, **interchange** the rows making sure that the resulting matrix is in echelon form.
7. Using **replacement** row operations, kill all non-zero entries in the column above the pivot entries.

1.2.1 Solution of Linear Systems

When we converted the augmented matrix of a linear system into its reduced row echelon form, we can write out the entire solution set of the system.

Example 1.15. Let

$$\begin{bmatrix} 1 & 0 & -5 & 1 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

be the augmented matrix of a linear system; then the system is equivalent to

$$\begin{array}{rcl} x_1 & -5x_3 & = 1 \\ x_2 + x_3 & = 4 \\ 0 & = 0 \end{array}$$

The variables x_1 and x_2 correspond to pivot columns in the matrix and are called **basic variables** (also **leading** or **pivot variables**). The other variable, x_3 is a **free variable**.

Free variables can be assigned arbitrary values and then leading variables expressed in terms of free variables:

$$\begin{aligned}x_1 &= 1 + 5x_3 \\x_2 &= 4 - x_3 \\x_3 &\text{ is free}\end{aligned}$$

Theorem 1.16 (Existence and Uniqueness). A linear system is consistent if and only if the rightmost column of the augmented matrix is **not** a pivot column—that is, if and only if an echelon form of the augmented matrix has no row of the form

$$\begin{bmatrix} 0 & \cdots & 0 & b \end{bmatrix} \text{ with } b \text{ nonzero}$$

If a linear system is consistent, then the solution set contains either

- (i) a unique solution, when there are no free variables, or
- (ii) infinitely many solutions, when there is at least one free variable.

Definition 1.17 (Using row reduction to solve a linear system).

1. Write the augmented matrix of the system.
2. Use the row reduction algorithm to obtain an equivalent augmented matrix in echelon form. Decide whether the system is consistent.
3. if the system is consistent, get the reduced echelon form.
4. Write the system of equations corresponding to the matrix obtained in Step 3.
5. Express each basic variable in terms of any free variables appearing in the equation.

Example 1.18 (System of Linear Equations - unique solution).

$$\begin{cases} -2x_1 - 4x_2 + 17x_3 = 20 \\ 10x_1 - 4x_2 + x_3 = 0 \\ 3x_1 + 5x_2 - 2x_3 = 14 \end{cases}$$

Solution

$$(x_1, x_2, x_3) = (1, 3, 2)$$

Example 1.19 (System of Linear Equations - unique solution, solved). Solve the following problem

$$\begin{cases} x_1 - 2x_2 + x_3 = 0 \\ 2x_2 - 8x_3 = 8 \\ -4x_1 + 5x_2 + 9x_3 = -9 \end{cases}$$

Solution:

$$\begin{array}{rcl} x_1 & - & 2x_2 + x_3 = 0 \\ & & 2x_2 - 8x_3 = 8 \\ -4x_1 & + & 5x_2 + 9x_3 = -9 \end{array} \quad \left[\begin{array}{cccc} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ -4 & 5 & 9 & -9 \end{array} \right]$$

$$\begin{array}{rcl} x_1 & - & 2x_2 + x_3 = 0 \\ & & 2x_2 - 8x_3 = 8 \\ & - & 3x_2 + 13x_3 = -9 \end{array} \quad \left[\begin{array}{cccc} 1 & -2 & 1 & 0 \\ 0 & 2 & -8 & 8 \\ 0 & -3 & 13 & -9 \end{array} \right]$$

$$\begin{array}{rcl} x_1 & - & 2x_2 + x_3 = 0 \\ & & x_2 - 4x_3 = 4 \\ & - & 3x_2 + 13x_3 = -9 \end{array} \quad \left[\begin{array}{cccc} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & -3 & 13 & -9 \end{array} \right]$$

$$\begin{array}{rcl} x_1 & - & 2x_2 + x_3 = 0 \\ & & x_2 - 4x_3 = 4 \\ & & x_3 = 3 \end{array} \quad \left[\begin{array}{cccc} 1 & -2 & 1 & 0 \\ 0 & 1 & -4 & 4 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

$$\begin{array}{rcl} x_1 & - & 2x_2 = -3 \\ & & x_2 = 16 \\ & & x_3 = 3 \end{array} \quad \left[\begin{array}{cccc} 1 & -2 & 0 & -3 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

$$\begin{array}{rcl} x_1 & & = 29 \\ & x_2 & = 16 \\ & x_3 & = 3 \end{array} \quad \left[\begin{array}{cccc} 1 & 0 & 0 & 29 \\ 0 & 1 & 0 & 16 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

Solution: (29, 16, 3)

Check: Is (29, 16, 3) a solution of the *original* system?

$$\begin{array}{rcl} x_1 & - & 2x_2 + x_3 = 0 \\ & & 2x_2 - 8x_3 = 8 \\ -4x_1 & + & 5x_2 + 9x_3 = -9 \end{array}$$

$$\begin{array}{rcl} (29) - 2(16) + 3 & = & 29 - 32 + 3 = 0 \\ 2(16) - 8(3) & = & 32 - 24 = 8 \\ -4(29) + 5(16) + 9(3) & = & -116 + 80 + 27 = -9 \end{array}$$

Example 1.20 (System of Linear Equations - no solution).

$$\left[\begin{array}{ccc|c} -2 & 3 & 1 & 3 \\ 11 & 5 & 7 & 4 \\ 1 & 20 & 12 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & \frac{16}{43} & 0 \\ 0 & 1 & \frac{25}{43} & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

Example 1.21 (System of Linear Equations - ∞ solutions).

$$\left[\begin{array}{ccc|c} -2 & 3 & 1 & 3 \\ 11 & 5 & 7 & 4 \\ 1 & 20 & 12 & 19 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & \frac{16}{43} & -\frac{3}{43} \\ 0 & 1 & \frac{25}{43} & \frac{41}{43} \\ 0 & 0 & 0 & 0 \end{array} \right]$$

Example 1.22. Solve the following problem

$$\begin{cases} x - y + 5z - 3w + 4t = -5 \\ 2x - y + 13z + 6w - 2t = -1 \\ x - 2y + 2z - 15w + 14t = -14 \\ 7x - 5y + 41z + 7w + 20t = -29 \\ -5x + 2y - 34z - 25w - 2t = 10 \end{cases}$$

$$\begin{aligned} & \left[\begin{array}{ccccc|c} 1 & -1 & 5 & -3 & 4 & -5 \\ 2 & -1 & 13 & 6 & -2 & -1 \\ 1 & -2 & 2 & -15 & 14 & -14 \\ 7 & -5 & 41 & 7 & 20 & -29 \\ -5 & 2 & -34 & -25 & -2 & 10 \end{array} \right] \xrightarrow{\begin{matrix} R_2 \leftarrow R_2 - 2R_1 \\ R_3 \leftarrow R_3 - R_1 \\ R_4 \leftarrow R_4 - 7R_1 \\ R_5 \leftarrow R_5 + 5R_1 \end{matrix}} \left[\begin{array}{ccccc|c} 1 & -1 & 5 & -3 & 4 & -5 \\ 0 & 1 & 3 & 12 & -10 & 9 \\ 0 & -1 & -3 & -12 & 10 & -9 \\ 0 & 2 & 6 & 28 & -8 & 6 \\ 0 & -3 & 9 & -40 & 18 & -15 \end{array} \right] \xrightarrow{\begin{matrix} R_1 \leftarrow R_1 + R_2 \\ R_3 \leftarrow R_3 + R_2 \\ R_4 \leftarrow R_4 - 2R_2 \\ R_5 \leftarrow R_5 + 3R_2 \end{matrix}} \\ & \left[\begin{array}{ccccc|c} 1 & 0 & 8 & 9 & -6 & 4 \\ 0 & 1 & 3 & 12 & -10 & 9 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 12 & -12 \\ 0 & 0 & 0 & -4 & -12 & -12 \end{array} \right] \xrightarrow{\begin{matrix} R_3 \leftrightarrow R_4/4 \\ R_5 \leftarrow R_5 + R_4 \end{matrix}} \left[\begin{array}{ccccc|c} 1 & 0 & 8 & 9 & -6 & 4 \\ 0 & 1 & 3 & 12 & -10 & 9 \\ 0 & 0 & 0 & 1 & 3 & -3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \xrightarrow{\begin{matrix} R_1 \leftrightarrow R_1 - 9R_3 \\ R_2 \leftarrow R_2 - 12R_3 \end{matrix}} \left[\begin{array}{ccccc|c} 1 & 0 & 8 & 0 & -33 & 31 \\ 0 & 1 & 3 & 0 & -46 & 45 \\ 0 & 0 & 0 & 1 & 3 & -3 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \end{aligned}$$

Thus, we arrived at

$$\begin{cases} x + 8z - 33t = 31 \\ y + 3z - 46t = 45 \\ w + 3t = -3 \end{cases}$$

solve it for x, y, w to get

$$\begin{cases} x = 31 - 8z + 33t \\ y = 45 - 3z + 46t \\ w = -3 \quad \quad - 3t \end{cases}$$

We may expression in future this solution as following

$$\begin{bmatrix} x \\ y \\ z \\ w \\ t \end{bmatrix} = \begin{bmatrix} 31 & - & 8s & + & 33p \\ 45 & - & 3s & + & 46p \\ & & s & & \\ -3 & & & - & 3p \\ & & & & p \end{bmatrix} = \begin{bmatrix} 31 \\ 45 \\ 0 \\ -3 \\ 0 \end{bmatrix} + \begin{bmatrix} -8 \\ -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} s + \begin{bmatrix} 33 \\ 46 \\ 0 \\ -3 \\ 1 \end{bmatrix} p$$

We don't have enough information to solve for all variables, therefore we have left with two, so called, free variables. This means we have infinitely many solution, a solution per each choice of s and p .

For example choose $s = 0$ and $p = 0$ for the solution is $(x, y, z, w, t) = (31, 45, 0, -3, 0)$, substitute it in first equations to get

$$x - y + 5z - 3w + 4t = 31 - 45 + 5 \cdot 0 + 9 + 0 + 0 = -5.$$

Another choice $s = -1/3$ and $p = -1$ gives $(x, y, z, w, t) = (2/3, 0, -1/3, 0, -1)$ with the same first equations gives

$$x - y + 5z - 3w + 4t = 2/3 - 0 - 5/3 - 3 \cdot 0 - 4 = -5$$

You can verify with other choices of z and t or more equations.

Example 1.23. For which value of a the following system has unique solution, no solution, infinitely many solutions.

$$\begin{cases} ax_1 + x_2 + x_3 = 1 \\ x_1 - ax_2 + x_3 = 1 \\ x_1 + x_2 - ax_3 = 1 \end{cases}$$

Solution: If $a = 1$ the system has only one equation $x + y + z = 1$ which has 2 free

variables, i.e. infinitely many solutions:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} s + \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} t$$

Assuming $a \neq 1$

$$\begin{aligned} & \left[\begin{array}{ccc|c} a & 1 & 1 & 1 \\ 1 & a & 1 & 1 \\ 1 & 1 & a & 1 \end{array} \right] \xrightarrow{R_1 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & a & 1 \\ a & 1 & 1 & 1 \\ 1 & a & 1 & 1 \end{array} \right] \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - aR_3} \left[\begin{array}{ccc|c} 1 & 1 & a & 1 \\ 0 & 1 - a^2 & 1 - a & 1 - a \\ 0 & a - 1 & 1 - a & 0 \end{array} \right] \xrightarrow[R_3 \leftarrow \frac{1}{a-1} R_3]{R_2 \leftarrow \frac{1}{1-a} R_2} \\ & \left[\begin{array}{ccc|c} 1 & 1 & a & 1 \\ 0 & 1 + a & 1 & 1 \\ 0 & 1 & -1 & 0 \end{array} \right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|c} 1 & 1 & a & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 1 + a & 1 & 1 \end{array} \right] \xrightarrow[R_3 \leftarrow R_3 - (1+a)R_2]{R_1 \leftarrow R_1 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & a + 1 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 2 + a & 1 \end{array} \right] \xrightarrow[\text{assume } a \neq -2]{R_3 \leftarrow \frac{1}{2+a} R_3} \\ & \left[\begin{array}{ccc|c} 1 & 0 & a + 1 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & \frac{1}{2+a} \end{array} \right] \xrightarrow[R_2 \leftarrow R_2 + R_3]{R_1 \leftarrow R_1 - (a+1)R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & \frac{1}{2+a} \\ 0 & 1 & 0 & \frac{1}{2+a} \\ 0 & 0 & 1 & \frac{1}{2+a} \end{array} \right] \end{aligned}$$

Thus, the system has a contradiction if $a = 2$ and so no solution. There is unique solution for every $a \neq -2, 1$ and as we found before infinitely many solutions for $a = 1$.

Example 1.24. For which value of a, b, c, d the following system has unique solution, no solution, infinitely many solutions.

$$\begin{cases} x - y + 3z = a \\ 2x + y + 3z = b \\ x + y + z = c \\ y - z = d \end{cases}$$

Solution

$$\left[\begin{array}{ccc|c} 1 & -1 & 3 & a \\ 2 & 1 & 3 & b \\ 1 & 1 & 1 & c \\ 0 & 1 & -1 & d \end{array} \right] \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & -1 & 3 & a \\ 0 & 3 & -3 & b - 2a \\ 0 & 2 & -2 & c - a \\ 0 & 1 & -1 & d \end{array} \right] \xrightarrow[R_3 \leftarrow R_3/2]{R_2 \leftarrow R_2/3} \left[\begin{array}{ccc|c} 1 & -1 & 3 & a \\ 0 & 1 & -1 & \frac{b-2a}{3} \\ 0 & 1 & -1 & \frac{c-a}{2} \\ 0 & 1 & -1 & d \end{array} \right]$$

Thus, in order the system be consistent it should satisfy

$$\frac{b-2a}{3} = \frac{c-a}{2} = d$$

1.3 Vector equations

A matrix with only one column is called a **column vector**, or simply a **vector**.

$\mathbb{R}^n$ is the set of all **column vectors** with n entries.

A **row vector**: a matrix with one row.

Two vectors are **equal** if and only if they have

- the same shape,
- the same number of rows,
- and their corresponding entries are equal.

The set of all vectors with n entries is denoted $\mathbb{R}^n$.

The **sum** $\mathbf{u} + \mathbf{v}$ of two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ is obtained by adding corresponding entries in $\mathbf{u}$ and $\mathbf{v}$. For example in $\mathbb{R}^2$

$$\begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The **scalar multiple** $c\mathbf{v}$ of a vector $\mathbf{v}$ and a real number (“**scalar**”) c is the vector obtained by multiplying each entry in $\mathbf{v}$ by c . For example in $\mathbb{R}^3$,

$$1.5 \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix} = \begin{bmatrix} 1.5 \\ 0 \\ -2 \end{bmatrix}.$$

The vector whose entries are all zeroes is called the zero vector and denoted $\mathbf{0}$:

$$\mathbf{0} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Operations with row vectors are defined in a similar way.

Example 1.25. Vector Equation 1

Consider

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} t + \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} s$$

express z as $z = ax + by + c$.

We have

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} t + \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} s = \begin{pmatrix} 1 + 2t + s \\ 2 + t + s \\ 3 + 5t + s \end{pmatrix}$$

Thus $x = 1 + 2t + s$, $y = 2 + t + s$ and $z = 3 + 5t + s$. We want to express

$$z = 3 + 5t + s = a(1 + 2t + s) + b(2 + t + s) + c = a + 2b + c + (a + b)s + (b + 2a)t$$

We arrive at

$$\begin{cases} -a + 2b + c = 3 \\ a + b = 1 \\ 2a + b = 5 \end{cases}$$

which solution $(a, b, c) = (4, -3, 4)$ therefore

$$z = 4x - 3y + 4$$

Example 1.26. Vector Equation 2 Consider $z = x + y - 3$, rewrite it in vector form.

$$(x, y, z) = (s, t, s + t - 3)$$

Definition 1.27 (Linear combinations). Given vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ in $\mathbb{R}^n$ and scalars $c_1, c_2, \dots, c_p$, the vector

$$\mathbf{y} = c_1 \mathbf{v}_1 + \dots + c_p \mathbf{v}_p$$

is called a **linear combination** of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p$ with **weights** $c_1, c_2, \dots, c_p$.

1.3.1 Rewriting a linear system as a vector equation

Consider an example: the linear system

$$\begin{aligned} x_2 + x_3 &= 2 \\ x_1 + x_2 + x_3 &= 3 \\ x_1 + x_2 - x_3 &= 2 \end{aligned}$$

can be written as equality of two vectors:

$$\begin{bmatrix} x_2 + x_3 \\ x_1 + x_2 + x_3 \\ x_1 + x_2 - x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 2 \end{bmatrix}$$

which is the same as

$$x_1 \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 2 \end{bmatrix}$$

Let us write the matrix

$$\begin{bmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -1 & 2 \end{bmatrix}$$

in a way that calls attention to its columns:

$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{b} \end{bmatrix}$$

Denote

$$\mathbf{a}_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \mathbf{a}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{a}_3 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

and

$$\mathbf{b} = \begin{bmatrix} 2 \\ 3 \\ 2 \end{bmatrix},$$

then the vector equation can be written as

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + x_3 \mathbf{a}_3 = \mathbf{b}.$$

Notice that to solve this equation is the same as express $\mathbf{b}$ as a linear combination of $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{a}_3$, and find all such expressions.

Therefore, *solving a linear system is the same as finding an expression of the vector of the right part of the system as a linear combination of columns in its matrix of coefficients.*

Theorem 1.28. A vector equation

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}.$$

has the same solution set as the linear system whose augmented matrix is

$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n & \mathbf{b} \end{bmatrix}$$

In particular $\mathbf{b}$ can be generated by a linear combination of $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ if and only if there is a solution of the corresponding linear system.

1.4 The matrix equation $A\mathbf{x} = \mathbf{b}$

Definition 1.29. If A is an $m \times n$ matrix, with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$, and if $\mathbf{x}$ is in $\mathbb{R}^n$, then the **product of A and $\mathbf{x}$** , denoted $A\mathbf{x}$, is the linear combination of the columns of A using the corresponding entries in $\mathbf{x}$ as weights:

$$\begin{aligned} A\mathbf{x} &= \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \\ &= x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n \end{aligned}$$

Example 1.30. The system

$$\begin{aligned} x_2 + x_3 &= 2 \\ x_1 + x_2 + x_3 &= 3 \\ x_1 + x_2 - x_3 &= 2 \end{aligned}$$

was written as

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + x_3\mathbf{a}_3 = \mathbf{b}.$$

where

$$\mathbf{a}_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \mathbf{a}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \mathbf{a}_3 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

and

$$\mathbf{b} = \begin{bmatrix} 2 \\ 3 \\ 2 \end{bmatrix}.$$

In the matrix product notation it becomes

$$\begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 2 \end{bmatrix}$$

or

$$A\mathbf{x} = \mathbf{b}$$

where

$$A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

Theorem 1.31. If A is an $m \times n$ matrix, with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$, and if $\mathbf{x}$ is in $\mathbb{R}^n$, the matrix equation

$$A\mathbf{x} = \mathbf{b}$$

has the same solution set as the vector equation

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$$

which has the same solution set as the system of linear equations whose augmented matrix is

$$\begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n & \mathbf{b} \end{bmatrix}.$$

Existence of solutions: The equation $A\mathbf{x} = \mathbf{b}$ has a solution if and only if $\mathbf{b}$ is a linear combination of columns of A .

Row-vector rule for computing $A\mathbf{x}$. If the product $A\mathbf{x}$ is defined then the i th entry in $A\mathbf{x}$ is the sum of products of corresponding entries from the row i of A and from the vector $\mathbf{x}$.

Theorem 1.4.5: Properties of the matrix-vector product $A\mathbf{x}$.

If A is an $m \times n$ matrix, $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, and c is a scalar, then

$$(a) \ A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v};$$

$$(b) \ A(c\mathbf{u}) = c(A\mathbf{u}).$$

Homogeneous linear systems: A linear system is **homogeneous** if it can be written as

$$A\mathbf{x} = \mathbf{0}.$$

A homogeneous system always has at least one solution $\mathbf{x} = \mathbf{0}$ (**trivial** solution).

Therefore for homogeneous systems an important question is existence of a **nontrivial** solution, that is, a nonzero vector $\mathbf{x}$ which satisfies $A\mathbf{x} = \mathbf{0}$:

The homogeneous system $A\mathbf{x} = \mathbf{b}$ has a nontrivial solution if and only if the equation has at least one free variable.

Example 1.32.

$$x_1 + 2x_2 - x_3 = 0$$

$$x_1 + 3x_3 + x_3 = 0$$

1.5 Solution sets of linear equations

Nonhomogeneous systems

When a nonhomogeneous system has many solutions, the general solution can be written in parametric vector form as one vector plus an arbitrary linear combination of vectors that satisfy the corresponding homogeneous system.

Example 1.33.

$$x_1 + 2x_2 - x_3 = 0$$

$$x_1 + 3x_3 + x_3 = 5$$

Theorem 1.34. Suppose the equation

$$A\mathbf{x} = \mathbf{b}$$

is consistent for some given $\mathbf{b}$, and $\mathbf{p}$ be a solution. Then the solution set of $A\mathbf{x} = \mathbf{b}$ is the set of all vectors of the form

$$\mathbf{w} = \mathbf{p} + \mathbf{v}_h,$$

where $\mathbf{v}_h$ is any solution of the homogeneous equation

$$A\mathbf{x} = \mathbf{0}.$$

2 Matrix operations

2.1 Matrix Addition

Labeling of matrix entries. Let A be an $m \times n$ matrix.

$$\begin{aligned} A &= \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix} \\ &= \begin{bmatrix} a_{11} & \cdots & a_{1j} & \cdots & a_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{i1} & \cdots & a_{ij} & \cdots & a_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} & \cdots & a_{mj} & \cdots & a_{mn} \end{bmatrix} \end{aligned}$$

Notice that in a_{ij} the first subscript i denotes the row number, the second subscript j the column number of the entry a_{ij} . In particular, the column $\mathbf{a}_j$ is

$$\mathbf{a}_j = \begin{bmatrix} a_{1j} \\ \vdots \\ a_{ij} \\ \vdots \\ a_{mj} \end{bmatrix}.$$

Diagonal matrices, zero matrices. The **diagonal entries** in A are

$$a_{11}, a_{22}, a_{33}, \dots$$

For example, the diagonal entries of the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

are 1, 5, and 9.

A **square matrix** is a matrix with equal numbers of rows and columns.

A **diagonal matrix** is a square matrix whose non-diagonal entries are zeroes.

Matrices

$$\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \quad \begin{bmatrix} \pi & 0 & 0 \\ 0 & \sqrt{2} & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

are all diagonal. The identity matrices

$$\begin{bmatrix} 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \dots$$

are diagonal.

Zero matrix. By definition, 0 is a $m \times n$ matrix whose entries are all zero. For example, matrices

$$\begin{bmatrix} 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

are zero matrices. Notice that zero square matrices, like

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

are diagonal!

Sums. If

$$A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}$$

and

$$B = \begin{bmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \end{bmatrix}$$

are $m \times n$ matrices then we define the sum $A + B$ as

$$\begin{aligned}
A + B &= \begin{bmatrix} \mathbf{a}_1 + \mathbf{b}_1 & \mathbf{a}_2 + \mathbf{b}_2 & \cdots & \mathbf{a}_n + \mathbf{b}_n \end{bmatrix} \\
&= \begin{bmatrix} a_{11} + b_{11} & \cdots & a_{1j} + b_{1j} & \cdots & a_{1n} + b_{1n} \\ \vdots & & \vdots & & \vdots \\ a_{i1} + b_{i1} & \cdots & a_{ij} + b_{ij} & \cdots & a_{in} + b_{in} \\ \vdots & & \vdots & & \vdots \\ a_{m1} + b_{m1} & \cdots & a_{mj} + b_{mj} & \cdots & a_{mn} + b_{mn} \end{bmatrix}
\end{aligned}$$

Scalar multiple. If c is a scalar then we define

$$\begin{aligned}
cA &= \begin{bmatrix} c\mathbf{a}_1 & c\mathbf{a}_2 & \cdots & c\mathbf{a}_n \end{bmatrix} \\
&= \begin{bmatrix} ca_{11} & \cdots & ca_{1j} & \cdots & ca_{1n} \\ \vdots & & \vdots & & \vdots \\ ca_{i1} & \cdots & ca_{ij} & \cdots & ca_{in} \\ \vdots & & \vdots & & \vdots \\ ca_{m1} & \cdots & ca_{mj} & \cdots & ca_{mn} \end{bmatrix}
\end{aligned}$$

Theorem 2.1.1: Properties of matrix addition. Let A , B , and C be matrices of the same size and r and s be scalars.

1. $A + B = B + A$
2. $(A + B) + C = A + (B + C)$
3. $A + 0 = A$
4. $r(A + B) = rA + rB$
5. $(r + s)A = rA + sA$
6. $r(sA) = (rs)A$.

2.2 Matrix multiplication

Multiplication of matrices. To answer the above question, we need to compute $A(B\mathbf{x})$ in matrix form.

Write $\mathbf{x}$ as

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

and observe

$$B\mathbf{x} = x_1\mathbf{b}_1 + \cdots + x_n\mathbf{b}_n.$$

Hence

$$\begin{aligned} A(B\mathbf{x}) &= A(x_1\mathbf{b}_1 + \cdots + x_n\mathbf{b}_n) \\ &= A(x_1\mathbf{b}_1) + \cdots + A(x_n\mathbf{b}_n) \\ &= x_1A(\mathbf{b}_1) + \cdots + x_nA(\mathbf{b}_n) \\ &= \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_n \end{bmatrix} \mathbf{x} \end{aligned}$$

Therefore multiplication by the matrix

$$C = \begin{bmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_n \end{bmatrix}$$

transforms $\mathbf{x}$ into $A(B\mathbf{x})$. Hence C is the matrix of the linear transformation $S \circ T$ and it will be natural to call C the product of A and B and denote

$$C = A \cdot B$$

(but the multiplication symbol “ $\cdot$ ” is frequently skipped).

Definition: Matrix multiplication. If A is an $p \times m$ matrix and B is an $m \times n$ matrix with columns

$$\mathbf{b}_1, \dots, \mathbf{b}_n$$

then the **product** AB is the $p \times n$ matrix whose columns are

$$A\mathbf{b}_1, \dots, A\mathbf{b}_n :$$

$$\begin{aligned} AB &= A \begin{bmatrix} \mathbf{b}_1 & \cdots & \mathbf{b}_n \end{bmatrix} \\ &= \begin{bmatrix} A\mathbf{b}_1 & \cdots & A\mathbf{b}_n \end{bmatrix}. \end{aligned}$$

Columns of AB . Each column $A\mathbf{b}_j$ of AB is a linear combination of columns of A with weights taken from the j th column of B :

$$\begin{aligned} A\mathbf{b}_j &= \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_m \end{bmatrix} \begin{bmatrix} b_{1j} \\ \vdots \\ b_{mj} \end{bmatrix} \\ &= b_{1j}\mathbf{a}_1 + \cdots + b_{mj}\mathbf{a}_m \end{aligned}$$

Mnemonic rules

$$[m \times n \text{ matrix}] \cdot [n \times p \text{ matrix}] = [m \times p \text{ matrix}]$$

$$\text{column}_j(AB) = A \cdot \text{column}_j(B)$$

$$\text{row}_i(AB) = \text{row}_i(A) \cdot B$$

Theorem 2.1.2: Properties of matrix multiplication. Let A be an $m \times n$ matrix and let B and C be matrices for which indicated sums and products are defined. Then the following identities are true:

1. $A(BC) = (AB)C$
2. $A(B + C) = AB + AC$
3. $(B + C)A = BA + CA$
4. $r(AB) = (rA)B = A(rB)$ for any scalar r
5. $I_m A = A = A I_n$

Powers of matrix. As it is usual in algebra, we define, for a square matrix A ,

$$A^k = A \cdots A \quad (k \text{ times})$$

If $A \neq 0$ then we set

$$A^0 = I$$

The transpose of a matrix. The **transpose** A^T of an $m \times n$ matrix A is the $n \times m$ matrix whose rows are formed from corresponding columns of A :

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

Theorem 2.1.3: Properties of transpose. Let A and B denote matrices whose sizes are appropriate for the following sums and products. Then we have:

1. $(A^T)^T = A$
2. $(A + B)^T = A^T + B^T$
3. $(rA)^T = r(A^T)$ for any scalar r
4. $(AB)^T = B^T A^T$

2.3 The inverse of a matrix

Invertible matrices

An $n \times n$ matrix A is **invertible** if there is an $n \times n$ matrix C such that

$$CA = I \text{ and } AC = I$$

C is called the **inverse** of A .

The inverse of A , if exists, is unique (!) and is denoted A^{-1} :

$$A^{-1}A = I \text{ and } AA^{-1} = I.$$

Singular matrices. A non-invertible matrix is called a **singular** matrix.

An invertible matrix is **nonsingular**.

Theorem 2.2.4: Inverse of a 2×2 matrix. Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

If $ad - bc \neq 0$ then A is invertible and

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

The quantity $ad - bc$ is called the **determinant** of A :

$$\det A = ad - bc$$

Theorem 2.2.5: Solving matrix equations. If A is an invertible $n \times n$ matrix, then for each $\mathbf{b} \in \mathbb{R}^n$, the equation $A\mathbf{x} = \mathbf{b}$ has the unique solution

$$\mathbf{x} = A^{-1}\mathbf{b}.$$

Theorem 2.2.6: Properties of invertible matrices.

(a) If A is an invertible matrix, then A^{-1} is also invertible and

$$(A^{-1})^{-1} = A$$

(b) If A and B are $n \times n$ invertible matrices, then so is AB , and

$$(AB)^{-1} = B^{-1}A^{-1}$$

(c) If A is an invertible matrix, then so is A^T , and

$$(A^T)^{-1} = (A^{-1})^T$$

2.4 Matrix determinant

The determinant $\det A$ of a square matrix A is a certain number assigned to the matrix; it is defined **recursively**, that is, we define first determinants of matrices of sizes 1×1 , 2×2 , and 3×3 , and then supply a formula which expresses determinants of $n \times n$ matrices in terms of determinants of $(n - 1) \times (n - 1)$ matrices.

The determinant of a 1×1 matrix $A = \begin{bmatrix} a_{11} \end{bmatrix}$ is defined simply as being equal its

only entry a_{11} :

$$\det \begin{bmatrix} a_{11} \end{bmatrix} = a_{11}.$$

The determinant of a 2×2 matrix is defined by the formula

$$\det \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} = a_{11}a_{22} - a_{12}a_{21}.$$

The determinant of a 3×3 matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

is defined by the formula

$$\det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = a_{11} \cdot \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix} - a_{12} \cdot \begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix} + a_{13} \cdot \begin{bmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

Example. The determinant $\det A$ of the matrix

$$A = \begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & 0 & 4 \end{bmatrix}$$

equals

$$2 \cdot \det \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix} + 7 \cdot \det \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

which further simplifies as

$$2 \cdot 4 + 7 \cdot (-1) = 8 - 7 = 1.$$

Submatrices. By definition, the submatrix A_{ij} is obtained from the matrix A by crossing out row i and column j .

For example, if

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix},$$

then

$$A_{22} = \begin{bmatrix} 1 & 3 \\ 7 & 9 \end{bmatrix}, \quad A_{31} = \begin{bmatrix} 2 & 3 \\ 5 & 6 \end{bmatrix}$$

Theorem 3.1.2: The determinant of a triangular matrix. If A is a triangular $n \times n$ matrix then $\det A$ is the product of the diagonal entries of A .

Corollary. The determinant of a diagonal matrix equals is the product of its diagonal elements:

$$\det \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & \cdots & & d_n \end{bmatrix} = d_1 d_2 \cdots d_n.$$

Corollary. The determinant of the identity matrix equals 1:

$$\det I_n = 1.$$

Corollary. The determinant of the zero matrix equals 0:

$$\det 0 = 0.$$

3 Cramer's Rule

Cramer's Rule is an explicit (closed) formula for solving systems of n linear equations with n unknowns and nonsingular (invertible) matrix of coefficients. It has important theoretical value, but is unsuitable for practical application.

For any $n \times n$ matrix A and any $\mathbf{b} \in \mathbb{R}^n$, denote by $A_i(\mathbf{b})$ the matrix obtained from A by replacing column i by the vector $\mathbf{b}$:

$$A_i(\mathbf{b}) = \begin{bmatrix} \mathbf{a}_1 & \cdots & \mathbf{a}_{i-1} & \mathbf{b} & \mathbf{a}_{i+1} & \cdots & \mathbf{a}_n \end{bmatrix}.$$

Theorem 3.3.7: Cramer's Rule. Let A be an invertible $n \times n$ matrix. For any

$\mathbf{b} \in \mathbb{R}^n$, the unique solution $\mathbf{x}$ of the linear system

$$A\mathbf{x} = \mathbf{b}$$

is given by

$$x_i = \frac{\det A_i(\mathbf{b})}{\det A}, \quad i = 1, 2, \dots, n.$$

For example, for a system

$$x_1 + x_2 = 3$$

$$x_1 - x_2 = 1$$

Cramer's rule gives the answer

$$x_1 = \frac{\det \begin{bmatrix} 3 & 1 \\ 1 & -1 \end{bmatrix}}{\det \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}} = \frac{-4}{-2} = 2$$

$$x_2 = \frac{\det \begin{bmatrix} 1 & 3 \\ 1 & 1 \end{bmatrix}}{\det \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}} = \frac{-2}{-2} = 1$$

4 Eigenvalues and eigenvectors

Quiz

What is the value of this determinant?

$$\Delta = \det \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 1 & 2 & 6 \end{bmatrix}$$

Quiz

Is Γ positive, negative, or zero?

$$\Gamma = \det \begin{bmatrix} 0 & 0 & 71 \\ 0 & 93 & 0 \\ 87 & 0 & 0 \end{bmatrix}$$

Eigenvectors. An **eigenvector** of an $n \times n$ matrix A is a nonzero vector $\mathbf{x}$ such that

$$A\mathbf{x} = \lambda\mathbf{x}$$

for some scalar λ .

Eigenvalues. A scalar λ is called an **eigenvalue** of A if there is a non-trivial solution $\mathbf{x}$ of

$$A\mathbf{x} = \lambda\mathbf{x};$$

$\mathbf{x}$ is called an **eigenvector** corresponding to λ .

Eigenvalue. A scalar λ is an eigenvalue of A **iff** the equation

$$(A - \lambda I)\mathbf{x} = \mathbf{0}$$

has a non-trivial solution.

Eigenspace. The set of all solutions of

$$(A - \lambda I)\mathbf{x} = \mathbf{0}$$

is a subspace of $\mathbb{R}^n$ called the **eigenspace** of A corresponding to the eigenvalue λ .

Theorem: Linear independence of eigenvectors. If $\mathbf{v}_1, \dots, \mathbf{v}_p$ are eigenvectors that correspond to distinct eigenvalues of $\lambda_1, \dots, \lambda_p$ of an $n \times n$ matrix A , then the set

$$\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$$

is linearly independent.

4.1 How to find eigenvectors and eigenvalues

Characteristic polynomial

The polynomial

$$\det(A - \lambda I)$$

in variable λ is **characteristic polynomial** of A .

For example, in the 2×2 case

$$\det \begin{bmatrix} a - \lambda & b \\ c & d - \lambda \end{bmatrix} = \lambda^2 - (a + d)\lambda + (ad - bc).$$

$$\det(A - \lambda I) = 0$$

is the **characteristic equation** for A .

Characterisation of eigenvalues

Theorem. A scalar λ is an eigenvalue of an $n \times n$ matrix A if and only if λ satisfies the characteristic equation

$$\det(A - \lambda I) = 0$$

Zero as an eigenvalue. $\lambda = 0$ is an eigenvalue of A if and only if $\det A = 0$.

Theorem: Eigenvalues of a triangular matrix. The eigenvalues of a triangular matrix are the entries on its main diagonal.

Proof. This immediately follows from the fact the the determinant of a triangular matrices is the product of its diagonal elements. Therefore, for example, for the 3×3 triangular matrix

$$A = \begin{bmatrix} d_1 & a & b \\ 0 & d_2 & c \\ 0 & 0 & d_3 \end{bmatrix}$$

its characteristic polynomial $\det(A - \lambda I)$ equals

$$\det \begin{bmatrix} d_1 - \lambda & a & b \\ 0 & d_2 - \lambda & c \\ 0 & 0 & d_3 - \lambda \end{bmatrix}$$

and therefore

$$\det(A - \lambda I) = (d_1 - \lambda)(d_2 - \lambda)(d_3 - \lambda)$$

has roots (eigenvalues of A)

$$\lambda = d_1, d_2, d_3.$$